$(\iota = \text{“}r = \textbf{call i8*}\ \textbf{malloc}(\textbf{i64}\ \mathit{len})\text{”})$
MALLOC

$$\frac{n = [\![\mathit{len}]\!]_R \quad c = \mathbf{i}(8 \times n) \Downarrow (\textbf{poison}) \quad P \text{ unallocated physical addresses,} \quad l \text{ fresh} \quad m' = m[l \mapsto (\mathsf{heap}, (\tau_{cur}, \infty), n, a, c, P)]}{R, (\tau_{cur}, m, C) \overset{\iota}{\hookrightarrow} R[r \mapsto \mathsf{Log}(l, 0, 0)], (\tau_{cur} + 1, m', C)}$$

$(\iota = \text{“}\textbf{call void free}(\textbf{i8*}\ \mathit{ptr})\text{”})$
FREE-LOGICAL

$$\frac{\mathsf{Log}(l, 0, 0) = [\![\mathit{ptr}]\!]_R \quad m(l) = (\mathsf{heap}, (b, \infty), n, a, c, P) \quad m' = m[l \mapsto (\mathsf{heap}, (b, \tau_{cur}), n, a, c, P)]}{R, (\tau_{cur}, m, C) \overset{\iota}{\hookrightarrow} R, (\tau_{cur} + 1, m', C)}$$

$(\iota = \text{“}r = \textbf{ptrtoint}\ \mathit{ty*}\ \mathit{op}\ \textbf{to i}\mathit{sz}\text{”})$
PTRTOINT-LOGICAL

$$\frac{\mathsf{Log}(l, o, s) = [\![\mathit{op}]\!]_R \quad \mathsf{cast2int}_M(l, o, s) = j}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto j \% 2^{sz}], M}$$

$(\iota = \text{“}r = \textbf{inttoptr i}64\ \mathit{op}\ \textbf{to}\ \mathit{ty*}\text{”})$
INTTOPTR

$$\frac{i = [\![\mathit{op}]\!]_R \quad p = \mathsf{Phy}(i, 0, \emptyset, \textbf{None})}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto p], M}$$

$(\iota = \text{“}r = \textbf{icmp eq}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-PTR-LOGICAL

$$\frac{\mathsf{Log}(l, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l, o_2, s) = [\![\mathit{op_2}]\!]_R}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto (o_1 = o_2)], M}$$

$(\iota = \text{“}r = \textbf{icmp eq}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-PTR-LOGICAL'

$$\frac{\mathsf{Log}(l_1, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l_2, o_2, s) = [\![\mathit{op_2}]\!]_R \quad l_1 \neq l_2}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \textbf{false}], M}$$

$(\iota = \text{“}r = \textbf{icmp eq}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-PTR-PHYSICAL

$$\frac{\mathsf{Phy}(o_1, s, I_1, cid_1) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Phy}(o_2, s, I_2, cid_2) = [\![\mathit{op_2}]\!]_R}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto (o_1 = o_2)], M}$$

$(\iota = \text{“}r = \textbf{icmp ule}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-ULE-PTR-PHYSICAL

$$\frac{\mathsf{Phy}(o_1, s, I_1, cid_1) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Phy}(o_2, s, I_2, cid_2) = [\![\mathit{op_2}]\!]_R}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto (o_1 \leq_u o_2)], M}$$

$(\iota = \text{“}r = \textbf{icmp eq}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-PTR-LOGICAL-NONDET-TRUE

$$\frac{\mathsf{Log}(l_1, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l_2, o_2, s) = [\![\mathit{op_2}]\!]_R \quad m(l_1) = (t_1, r_1, n_1, a_1, c_1, P_1) \quad m(l_2) = (t_2, r_2, n_2, a_2, c_2, P_2) \quad l_1 \neq l_2 \quad \big((o_1 = n_1 \wedge o_2 = 0) \vee o_1 > n_1 \vee (o_1 = 0 \wedge o_2 = n_2) \vee o_2 > n_2 \vee r_1, r_2 \text{ disjoint}\big)}{R, (\tau_{cur}, m, C) \overset{\iota}{\hookrightarrow} R[r \mapsto \textbf{true}], (\tau_{cur}, m, C)}$$

$(\iota = \text{“}r = \textbf{icmp ule}\ \mathit{ty*}\ \mathit{op_1}\ \mathit{op_2}\text{”})$
ICMP-PTR-ULE-LOGICAL

$$\frac{\mathsf{Log}(l, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l, o_2, s) = [\![\mathit{op_2}]\!]_R}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto (o_1 \leq_u o_2)], M}$$

$(\iota = \text{“}r = \textbf{gep}\ \mathit{ty*}\ \mathit{op_1}\ \textbf{i}\mathit{sz}\ \mathit{op_2}\text{”})$
GEP-LOGICAL

$$\frac{\mathsf{Log}(l, o, s) = [\![\mathit{op_1}]\!]_R \quad i = [\![\mathit{op_2}]\!]_R \quad o' = (o + \mathsf{bytewidth}(ty) * i) \% 2^{\mathsf{ptrsz}(s)}}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \mathsf{Log}(l, o', s)], M}$$

$(\iota = \text{“}r = \textbf{gep}\ \mathit{ty*}\ \mathit{op_1}\ \textbf{i}\mathit{sz}\ \mathit{op_2}\text{”})$
GEP-PHYSICAL

$$\frac{\mathsf{Phy}(o, s, I, cid) = [\![\mathit{op_1}]\!]_R \quad i = [\![\mathit{op_2}]\!]_R \quad o' = (o + \mathsf{bytewidth}(ty) * i) \% 2^{\mathsf{ptrsz}(s)}}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \mathsf{Phy}(o', s, I, cid)], M}$$

$(\iota = \text{“}r = \textbf{gep inbounds}\ \mathit{ty*}\ \mathit{op_1}\ \textbf{i}\mathit{sz}\ \mathit{op_2}\text{”})$
GEP-INBOUNDS-LOGICAL

$$\frac{\mathsf{Log}(l, o, s) = [\![\mathit{op_1}]\!]_R \quad o' = o + \mathsf{bytewidth}(ty) * i \quad i = [\![\mathit{op_2}]\!]_R \quad \mathsf{inbounds}_M(l, o) \quad \mathsf{inbounds}_M(l, o')}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \mathsf{Log}(l, o', s)], M}$$

$(\iota = \text{“}r = \textbf{gep inbounds}\ \mathit{ty*}\ \mathit{op_1}\ \textbf{i}\mathit{sz}\ \mathit{op_2}\text{”})$
GEP-INBOUNDS-PHYSICAL

$$\frac{\mathsf{Phy}(o, s, I, cid) = [\![\mathit{op_1}]\!]_R \quad i = [\![\mathit{op_2}]\!]_R \quad o' = o + \mathsf{bytewidth}(ty) * i \quad 0 \leq o' < 2^{\mathsf{ptrsz}(s)}}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \mathsf{Phy}(o', s, I \cup \{o, o'\}, cid)], M}$$

$(\iota = \text{“}r = \textbf{psub}\ \mathit{ty*}\ \mathit{op1},\ \mathit{op2}\text{”})$
PSUB-LOGICAL

$$\frac{\mathsf{Log}(l, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l, o_2, s) = [\![\mathit{op_2}]\!]_R \quad i = (o_1 - o_2) \% 2^{\mathsf{ptrsz}(s)}}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto i], M}$$

$(\iota = \text{“}r = \textbf{psub}\ \mathit{ty*}\ \mathit{op1},\ \mathit{op2}\text{”})$
PSUB-LOGICAL-POISON

$$\frac{\mathsf{Log}(l_1, o_1, s) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Log}(l_2, o_2, s) = [\![\mathit{op_2}]\!]_R \quad l_1 \neq l_2}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto \textbf{poison}], M}$$

$(\iota = \text{“}r = \textbf{psub}\ \mathit{ty*}\ \mathit{op1},\ \mathit{op2}\text{”})$
PSUB-PHYSICAL

$$\frac{\mathsf{Phy}(o_1, s, I_1, cid_1) = [\![\mathit{op_1}]\!]_R \quad \mathsf{Phy}(o_2, s, I_2, cid_2) = [\![\mathit{op_2}]\!]_R \quad i = (o_1 - o_2) \% 2^{\mathsf{ptrsz}(s)}}{R, M \overset{\iota}{\hookrightarrow} R[r \mapsto i], M}$$

Fig. 5. Selected rules of our operational semantics